



Giacomo Castagnetti

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Master student at ETHZ, building software tools to solve biological problems.

Work experience

Roche - Multimodal Machine Learning

Apr 2026 - Apr 2027

- Building an agnostic molecular encoder for single cell omics data.

ETH Zurich - De-novo Drug Design Internship

Nov 2025 – Mar 2026

- Setting up a precise **validation framework** that highlighted a failure mode of a drug design model.
- **Improved** training time by **20%** and **implemented** multiple successful **solutions** to the problem.
- Provided **visualizations** and reports for non-technical people on model limitations and next steps.

Biozentrum - Bioinformatics Internship

Jul 2023 – Sep 2023

- Selected to the Biozentrum Research Summer program, in [Richard Neher's](#) lab.
- Developed a **novel tool** for tracking recombination in bulk ONT sequencing data (10x faster than the traditional experimental pipeline) ([GitHub](#)).
- Tracked the evolution and spreading of West Nile Virus (**Nextstrain**) ([GitHub](#)).

Education

Master in Computational Biology and Bioinformatics ([link](#))

ETH Zurich, Switzerland

Combining rigorous training in computer science, mathematics, and biology.

GPA: 5.2 | 2024 - 2026

- Relevant coursework: Machine Learning, LLMs, Big Data, Mathematical Modelling.

Bachelor in Genomics ([link](#)) - graduation cum laude and honorable mention

University of Bologna, Italy

Highly interdisciplinary bachelor which gave me theoretical foundations in statistics, programming and molecular biology; together with practical bioinformatics skills.

GPA: 6 | 2021 - 2024

Publications

Automated bacteriophage evolution studies with the Aionostat. doi.org/10.1101/2025.04.16.649189

April 2025

Projects

Agentic Pipeline for Job Scoring

Mar 2026

Orchestrated multi-agent pipeline that scrapes job postings and indexes them against a provided resume.

Phagos Hackaton

Paris, France | May 2025

Used large protein and nucleotide foundation models to classify bacteriophage lifestyle.

President of student association CCBB

Sep 2025 - Sep 2026